

Supplementary Information

Measurement choices in longitudinal citation language comparisons around replication projects

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Scientometrics, Online Resource 1

This supplement accompanies the exploratory comparison of model-score definitions, literal word forms and text-retention choices. It retains source reconstruction, measurement and production diagnostics, every specified lexical family, exact selection decomposition and finite-corpus retention benchmarks. Semantic acknowledgment and target-specific updating are not inferred from these observables. Only aggregate legacy-rating diagnostics are presented; private annotation-export metadata and raw participant records are excluded.

S1 Source reconstruction and target criteria

The source flow is 141 target identifiers $\rightarrow$ 135 archived distinct-title records $\rightarrow$ 110 resolver returns $\rightarrow$ 108 title-audited original papers $\rightarrow$ the retained conservative 105-paper baseline. The six omitted same-paper target identifiers are ml1.4, ml1.5, ml1.6, ml1.16, rpp.149 and rpp.50. The two wrong-paper matches excluded after resolution are ee7 and rpp.117. Twenty-five seed records were unresolved by the historical resolver. These selection stages do not constitute an exhaustive census of each project’s target effects. Subsequent resolution of a selected statistic is distinguished from a unique source-wide interpretation; the named 106-source sensitivity below leaves the baseline unchanged. The resolver used title match and search routes, restricted candidates to source publication year ± 2 , chose the highest citation count, and rejected candidates below 20 citations. The final cutoff was a historical low-count mismatch heuristic, not an established identity-validity criterion. The surviving record does not provide verified, mutually exclusive reason counts for the 25 unresolved titles.

Table S1: Source reconstruction by project.

Project	Raw target IDs	Archived titles	Retained papers	Baseline papers
RPP	97	95	74	71
EERP	18	18	14	14
Many Labs 1	16	12	11	11
Many Labs 3	10	10	9	9
Total	141	135	108	105

Outcome assignment concerns the selected test, not every claim in the original article. For multi-laboratory projects, the revised ledger uses the relevant project aggregate and specified

criterion rather than a median across mixed laboratory and aggregate rows. Many Labs 3 denotes the whole aggregate, including MTurk where administered, rather than only campus samples. Every retained assignment is keyed to a source identifier and accompanied by its evidence and scope. Nominal $p < .05$ and the original direction or pattern define criterion-positive, retaining each report’s one- or two-sided test and requiring every component of an explicit joint rule. Weak aggregate support in Many Labs 1 and conceptual adaptations in Many Labs 3 are documented rather than promoted to universal successful replication. The three separately reported cases are never coded negative by default.

Three source cases were resolved by executable checks of public primary data in addition to report inspection. For rpp.52, the original linear-trend contrast is nonsignificant (two-sided $p=0.321966$), distinct from an omnibus interaction. For rpp.69, all six specified two-sided binomial tests show recognition above chance and $p<0.05$; the largest p is 6.60×10^{-7} . The latter raw file has 333 rows while the report states 330 participants; 324 nonmissing selected-target responses enter the tests. Duplicated participant-number codes are not silently resolved, and the provided sensitivity does not change the six-test decision. For rpp.93, the declared conditional simple effect remains negative under the criterion (p approximately 0.632), rather than substituting an archived three-way-interaction statistic. These checks are numerical source audits, not new human citation annotations.

Table S2: Separately reported criterion cases and analytic treatment.

Separate case	Verified result and unresolved interpretation	Treatment
rpp.19	Original target-specific contrast and archived statistic concern different comparisons; implementation/label conflicts remain	Exclude from retained baseline; enumerate alternative labels
rpp.73	Archived Staunton contrast is positive; no unique source-wide aggregation across sites is specified	Retain baseline exclusion; add named 106-source selected-contrast sensitivity
rpp.154	Self-report mean reproduced; primary nominal p is inapplicable and dated OR/AND interval rules conflict	Keep outside nominal-significance baseline; report rule-specific axes

For rpp.19, the reproduced archived statistic ($p=0.0708827$) compares stored listtype labels using each participant’s equal-weight mean across correct, intrusion and new-probe accuracy. Conflicts between surviving code, logged patterns and labels prevent treating it as a verified presentation-order effect. A report labeled pre-data collection and dated 19 September 2014 does not by that printed date establish prospective registration or the exact runtime version. The target remains unresolved and excluded from the retained baseline contrast.

For rpp.73, the public Staunton counts are 25 Democratic and 95 Republican bumper-sticker observations, against voter proportions 0.30/0.70. The unrounded Pearson $\chi^2(1)=4.801587$ gives $p=0.0284335$ in the predicted direction; the archived $p=0.0284597$ uses rounded $\chi^2=4.80$. An actual May 2013 registered protocol names two initial communities, both directionally positive, and two secondary communities; all four include one reverse-direction result. It supplies no unique AND/OR/pooling rule. The selected-site statistic is therefore resolved, while source-wide interpre-

tation remains an analytic choice. The full original Study 3 text was not independently obtained in this pass; original-target corroboration uses its registered quotation and replication materials. Public report, counts, registration and selected-site script are linked in the accompanying source clarification.

For rpp.154, newly recovered Materials files reproduce the primary aggregate self-report conscientiousness mean $r = -0.106296$ and the pairwise-matrix GDP-adjusted mean $r = -0.212127$, matching the report's -0.11 and table's -0.21 after rounding. These are not the distinct PNC construct. Previously audit-linked data differ, and the report's prose value -0.26 is not reproduced for the replication. Dated September 2014 wiki versions and an October registered snapshot retain an OR interval criterion, whereas the January 2015 report and later pre-data-labelled file use a joint AND criterion. Earlier wording also misidentifies the original unadjusted -0.43 as partial. Registration here preserves a dated state; it does not establish collection timing or resolve the criterion conflict.

Using the report's stated reference intervals without independently endorsing their construction, the unadjusted self-report mean is outside its reference interval and the adjusted mean is inside its matching interval. Thus the later AND rule is not met, while the matching-adjusted-interval interpretation of the earlier OR rule is met. A literal reading of its mislabeled target leaves the alternative overlap condition unresolved. No justified primary nominal significance test of this dependent-correlation average was established; not applicable is distinct from nonsignificant or missing. The source clarification retains construct, version, numeric result, rule history, author narrative and analytic inclusion on separate axes. These findings do not supply a new universally applicable inclusion rule for the other 105 papers.

The exact 108-row source ledger and attrition records are accompanying machine-readable artifacts. Title matching validates paper identity only. The archive does not independently adjudicate whether each later text restates the selected replicated proposition.

S2 Dates and retrieval scope

Table S3: Typed project-publication milestones and timing sensitivities.

Project	Primary journal year	Verified publication milestone	Recorded earlier manuscript date
RPP	2015	Journal publication 28 August 2015	Individual target evidence can predate the project
EERP	2016	Online 3 March 2016	No exhaustive first-availability claim
Many Labs 1	2014	Online 19 May 2014	OSF upload 25 November 2013; historical bytes not certified
Many Labs 3	2016	November 2016 issue; online date unresolved	OSF upload 9 March 2015; historical bytes not certified

Journal years define the main event alignment. Recorded-upload years define a sensitivity with RPP/EERP/ML1/ML3 years 2015/2016/2013/2015. Public-state and upload logs establish recorded events, not the exact content of an unavailable historical file or author awareness. One

earlier individual publication is separately documented; substituting it where known does not reconstruct all earlier evidence.

Retrieval occurred in July 2026 through Semantic Scholar. Requests contained 100 citation records and targeted approximately 300 contexts per source. For sources with at most 1,500 citations, starting offsets were 0, 100 and 200. Larger sources used three deterministically and evenly placed, rounded offsets from zero to $\min(\text{citation count} - 100, 9,000)$. The 9,000 cap was the historical implementation’s conservative starting offset within a 10,000-record window, not verification of a current endpoint guarantee. The code retained contexts whose original length was greater than 40, stripped whitespace, and initially deduplicated by (citing year, first 80 stripped characters). Distinct continuations sharing a prefix may therefore have been merged before the later exact-string stage. Missing citing-paper/version identifiers prevent treating the archive as a census of independent articles.

The audited source archive contains 15,544 retrieval records, of which 15,535 have assertion-filter outputs. Nine missing outputs are not negatives. Exact within-source deduplication removes 144 scored records, yielding 15,391 text units and 8,480 positives before the three criterion exclusions. There are 142 same-text/multiple-year groups, including 83 positive groups; five have conflicting filter labels. Eighty-three retained texts are earlier than the source journal year. These are flags for review, not certified chronology errors. Thirteen texts before 1993 postdate their older original source papers; no arbitrary 1993 cutoff is applied.

Thirty exact-text groups occur under multiple source papers, comprising 63 rows across 24 sources. Resampling text-sharing components is a limited dependence check. It cannot identify unknown shared citing papers or repair missing version identifiers.

S3 Measurement sample and corrected agreement

The historical Task A key contains 368 presentations: 330 ordinary sentences, 26 quality-control items and 12 repeat presentations. Both accepted raters rated all presentations. Certainty is conditional on each rater’s assertion decision; an absent certainty rating for a nonassertion is structural missingness, not zero. Primary agreement retains the canonical ordinary record and excludes quality-control and duplicate-marked records. Five duplicate-marked records precede their canonical originals in task order; a first-presentation sensitivity is therefore also retained.

The verbal score anchors were 0, fully tentative or strongly conditional; 0.25, cautious suggestion; 0.5, neutral or distanced report; 0.75, strong evidential statement; and 1, categorical, causal, axiomatic or unqualified general-present assertion. The legacy human manual normally capped explicit failed-replication or challenged-evidence language at 0.25 while allowing categorical negative restatements a score of 1. The archived model prompt retained both rules. This is a shared operational rubric rather than evidence of an instruction violation; additional ratings under it would not alone establish a monotonic measure of commitment to a replicated proposition.

Table S4: Human measurement sample and corrected agreement.

Items	Assertion n	κ and 95% interval	Paired- positive n	Ordinal α and 95% interval
All presentations	368	0.654 [0.551, 0.748]	274	0.603 [0.507, 0.687]
Unique including QC	356	0.671 [0.572, 0.760]	265	0.603 [0.510, 0.688]
Ordinary unique	330	0.653 [0.546, 0.752]	248	0.562 [0.459, 0.654]
Ordinary excluding flagged	327	0.660 [0.550, 0.757]	246	0.569 [0.463, 0.660]

Intervals use 9,999 sentence-cluster bootstrap draws, retaining both raters. When repeated presentations are included, presentations of the same sentence move together. Ordinal alpha uses pairable ratings and recomputes category-frequency distances in each draw. The first-task-presentation rule yields ordinary-item alpha 0.5566. These estimates condition on the sampled sentences and raters; they do not estimate scorer-selection or retrieval uncertainty. The ordinary alpha point estimate is below the archive’s own 0.60 gate; its interval crossing 0.60 does not establish that the gate was met. That gate is not a universal validity threshold.

The 330 ordinary items comprise 248 paired-positive, 46 paired-negative and 36 disagreement cases. The ordinary sample was drawn from model-positive texts under 70–350-character and readability restrictions. It cannot estimate false-negative filtering, specificity or accuracy over all retrieved texts. Among 261 exact-linked, human-consensus items, positive-label agreement is 84.3%, a conditional agreement statistic.

S4 Development agreement and production repeatability

Table S5: Development-sample model agreement.

Score run	Development ρ	MAE	Mean model minus human	Model on human slope
Former Qwen	0.307	0.2125	+0.0361	0.2415
Luna	0.662	0.1421	−0.0081	0.6858
SiliconFlow	0.621	0.1421	−0.0071	0.5938
Equal-weight mean	0.695	0.1215	−0.0076	0.6398

All four rows use the same 248 ordinary paired-positive items. The equal-weight score has ρ interval [0.609,0.771], MAE interval [0.1033,0.1401] and model-on-human slope interval [0.5473,0.7302]. Mean model scores at human means 0 and 1 are 0.2411 (n=14) and 0.8095 (n=21). Low overall mean bias therefore coexists with endpoint compression. The human reference is itself imperfect, and these ordinal coding anchors are not probability forecasts. These diagnostics do not identify a valid multiplier for substantive effects. The former Qwen scorer’s same-248 Spearman coefficient is 0.306729; the equal-weight revised score gives 0.695438 (Pearson 0.707484), meeting the protocol’s model–human $\rho \geq 0.60$ criterion. The located protocol does not make an untouched holdout a condition of that numerical criterion. This resolved model-agreement

criterion is separate from the ordinary human $\alpha = 0.562265$ and from construct validity. The obsolete 0.42 value belongs to the prior rejected version and is not the current score result; its exact 110-paper manuscript was not reconstructed from the later 4 October archive.

An exact source-and-text match links 220 development items to production scores. Their mean changed on 97 items (44.1%); between-run MAE is 0.0756 and mean production-minus-development difference is -0.0233 , with interval $[-0.0398, -0.0068]$. Four differences exceed 0.25. Production scores correlate with the human mean at $\rho = 0.722$ $[0.633, 0.794]$, with MAE 0.1250. These remain previously selected items, not an independent holdout. No raw annotation-export metadata are needed to interpret these aggregate facts, and none are included here.

Table S6: Frozen scoring error by criterion group and journal period.

Source criterion and journal period	Items	Sources	Mean frozen score minus human
Negative before	56	28	-0.0379
Negative at or after	71	31	+0.0106
Positive before	29	20	-0.0129
Positive at or after	63	34	-0.0377

For these 219 exact-linked items, the criterion-negative minus criterion-positive difference in post-minus-pre mean error is $+0.0733$ $[-0.0357, +0.1728]$. Sources are resampled within criterion groups, retaining their pre/post items together; cell means are sentence-weighted. Recorded-upload-year alignment gives $+0.093$ $[-0.019, +0.190]$. These selected-sample diagnostics do not establish measurement invariance or justify subtracting a bias correction from the main estimates.

S5 Frozen estimands and principal estimates

Source means require at least 15 positive texts. Lifetime slopes require three adaptive bins of at least eight texts, three distinct rounded bin years and a four-year bin-mean-year span. With text-count weights, γ is the slope of $\log(1 - \text{mean score})$ on mean year and the historical transformed drift is $1 - \exp(\gamma)$. The raw-score slope uses the same bins. Event changes require ten positive texts before and ten at or after the journal year. These eligibility rules define different conditional source samples.

Table S7: Baseline model-score endpoints and conditional uncertainty.

Endpoint	Negative / positive n	Difference	Bootstrap 95% interval	Welch 95% interval
Mean score	51 / 45	-0.0042	$[-0.0218, +0.0134]$	$[-0.0221, +0.0137]$
Transformed lifetime drift	45 / 40	-0.0005	$[-0.0072, +0.0063]$	$[-0.0075, +0.0064]$
Project-aligned change	40 / 32	+0.0031	$[-0.0240, +0.0299]$	$[-0.0247, +0.0308]$
Raw score slope per year	45 / 40	+0.0002	$[-0.0031, +0.0036]$	$[-0.0033, +0.0037]$

Table S7 (continued)

Endpoint	Negative / positive n	Difference	Bootstrap 95% interval	Welch 95% interval
Positive-filter fraction	57 / 48	+0.0792	[+0.0321, +0.1256]	[+0.0309, +0.1274]

Primary bootstrap intervals use 19,999 source draws within criterion groups. All fitted score values, source assignments, text retrieval and dates are held fixed. The equal-paper pre/post means are 0.4985/0.5070 for 40 criterion-negative sources and 0.5022/0.5076 for 32 criterion-positive sources. Their group changes are +0.0085 and +0.0054. A small difference between those changes does not imply no change within either group. Project results concern both groups; neither is an untreated control.

S6 Sensitivity and label uncertainty

The fixed post hoc diagnostic grid varies event timing, inclusion of the event year, conflicting-date handling, readability eligibility, scoring component, symmetric event windows, minimum text counts, partial retrieval year and project composition. Changing these choices can change eligibility. Differences between specifications do not isolate one causal mechanism.

Table S8: Alternative-label allocation ranges.

Endpoint	Allocations	Estimate range	Envelope of conditional intervals
level	8	-0.0055 to -0.0026	-0.0228 to +0.0148
transformed_drift	8	-0.0005 to -0.0005	-0.0072 to +0.0063
event_change	8	+0.0011 to +0.0055	-0.0263 to +0.0326

There are eight allocations but only four distinct level estimates, one drift estimate and two event estimates because some separately reported papers are ineligible for some endpoints. Duplicate values are not independent corroboration. The envelope is not a single calibrated 95% interval and does not include all target-link, retrieval, date or measurement uncertainty.

A separately named archived-Staunton-selected-contrast sensitivity adds rpp.73 as criterion-positive, yielding 106 papers (49 positive, 57 negative), 15,302 text units and 8,443 positives. This is a transparent alternative selected-contrast convention, not a retrospective assertion that the baseline criterion was always unique. Its source has 22 texts and 16 positives. Model-score level eligibility rises to 97 papers (51 negative, 46 positive), with contrast -0.003259 and source-bootstrap interval $[-0.020744, +0.014580]$. Lifetime drift is unchanged because 16 positives cannot fill three eight-text bins; model-score event estimates are unchanged because seven pre-event and nine post-event positives fail the ten-per-period rule. The explicit definition and all numeric endpoints are in `staunton_sensitivity_definition.json` and `staunton106_frozen_model_sensitivity.json`.

The pre-event raw-score slope comparison over years -8 to -1 includes 68 papers and is -0.015734 . Its bootstrap interval $[-0.031563, -0.000941]$ differs in inclusion of zero from its Welch interval $[-0.031555, +0.000087]$, with Welch $p=0.0512$. Neither this diagnostic nor a nonsignificant pretrend establishes parallel trends. Project-specific slopes are heterogeneous and use very small groups in some projects.

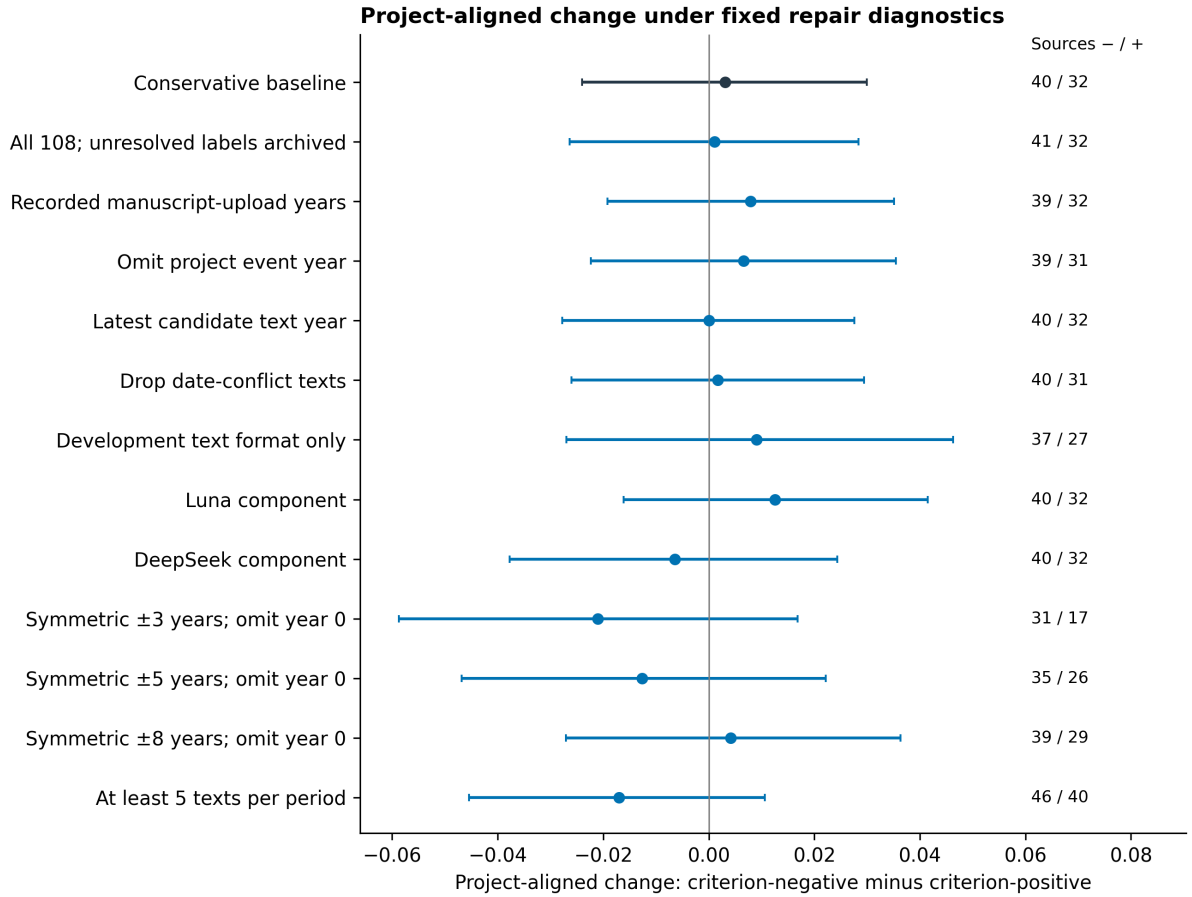


Fig. S1: Project-aligned change sensitivity figure. Bars are conditional 95% source-bootstrap intervals. The all-108 source-based row retains the separately reported cases under their archived labels; it is not an independently adjudicated full cohort. Timing alternatives do not certify true dates. The complete machine-readable grid accompanies the figure

S7 Precision and the meaning of equivalence

The supplied precision grid reports illustrative model-scale margins rather than a prespecified scientifically validated equivalence bound. At margin ± 0.02 , the level comparison passes the nominal TOST rule ($p=0.0417$), whereas project-aligned change does not ($p=0.1139$); the latter passes at ± 0.03 ($p=0.0285$). These arithmetic statements do not establish equivalence of human-rated certainty, belief updating or scientific correction. Measurement error and source uncertainty remain outside those tests. Hypothetical precision simulations condition on frozen cohort variances and are not achieved-power evidence.

S8 Archived acknowledgment task

Task B contains 150 selected acknowledgment items, distinct from the certainty task. Human-majority labels yield five positives, and four-model-majority labels yield seven. Their cross-tabulation has 142 shared negatives, four shared positives, three model-only positives and one human-only positive. Overall agreement of 97.3% accompanies positive predictive agreement of 4/7 and positive recall of 4/5 against an imperfect reference. Three-human Fleiss κ with categories 0,1 and unable-to-decide is 0.236 [0.036,0.433]. Treating unable-to-decide as missing

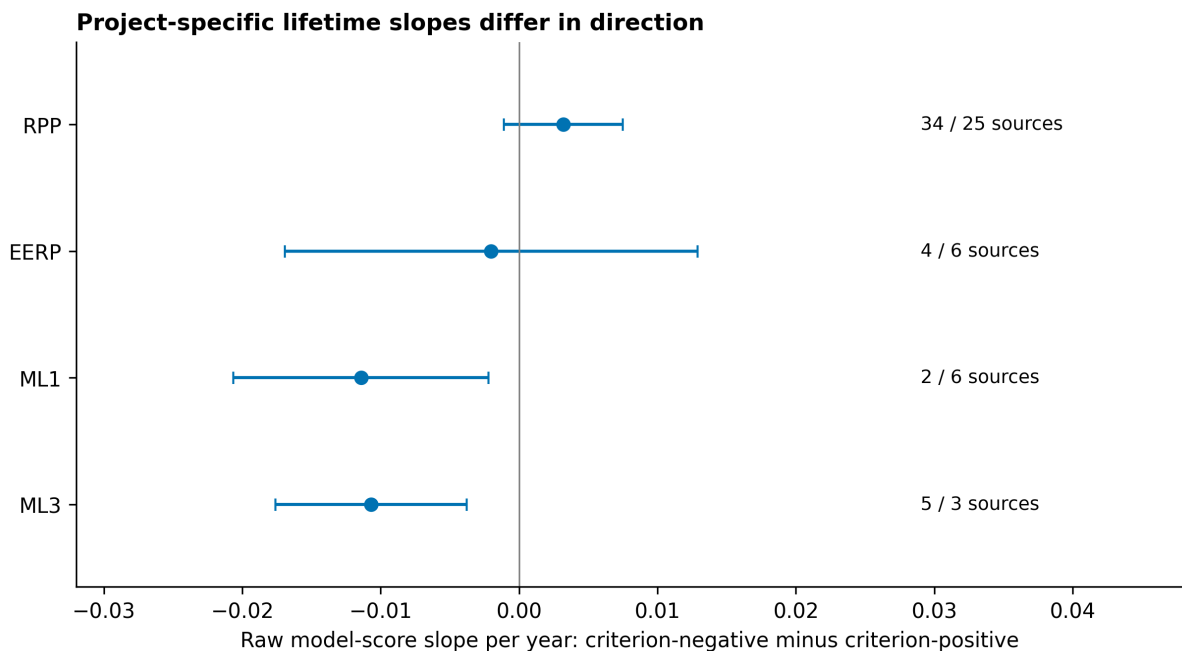


Fig. S2: Project-specific raw-score slope contrasts and Welch 95% intervals, using the same adaptive bins as the transformed drift. Group sizes are shown. These are post hoc descriptive heterogeneity checks, not independently established project-specific effects

gives a different estimand, nominal $\alpha=0.449$ [0.163,0.687].

The original sampling seed and exact frame are unavailable. Full-text linkage recovers 141 scored items; seven additional prefix matches do not establish full identity and two remain unmatched. The sample covers 27 source papers and cannot estimate population acknowledgment prevalence. Acknowledgment is partly encoded in the wording-score instructions, so its association with low scores is not independent construct validation.

S9 Literal vocabulary and conditional text denominators

This post hoc diagnostic counts visible forms in the retained 105-source corpus without new model calls or human labels. Before lexical counts and comparisons, the rule file fixed possibly, perhaps and probably, drawn as three examples from Kilicoglu and Bergler (2008, Table 1), and the separate forms replicate, replicates, replicated, replicating, replication and replications. These are not the source paper’s full lexicon or validated classifier. A sensitivity retains originally lowercase may, might and could, excluding numeric-adjacent may under a fixed ASCII-digit rule. The earlier score results and the archive’s broader dictionary were already known; this local freeze is not preregistration of the study.

Tokens are Unicode alphabetic sequences with at most one internal apostrophe, subject to the frozen larger-word/digit/underscore boundary guards. Core and replication forms are case-folded; curly apostrophes are normalized for matching, and a final possessive apostrophe-s is accepted for replication forms only. No stemming, synonym expansion, NFKC normalization, scope resolution or language classification is used. Modal forms must remain lowercase in the original text; the fixed numeric guard is not a date classifier. Every literal match is retained, including negation, quotation, unrelated targets and biological or computational senses. Replication-word presence

is neither necessary nor sufficient for acknowledgment, and its absence supplies no bound on acknowledgment prevalence.

Table S9: Every predefined literal form; full-text counts in the 105-source baseline.

Exact form	Hit text units	Occurrences	Hit-bearing sources	Qwen-positive / negative hits
possibly	24	24	23	19 / 5
perhaps	47	50	32	31 / 16
probably	16	16	14	12 / 4
replicate	85	86	37	46 / 39
replicates	13	13	12	7 / 6
replicated	93	95	48	45 / 48
replicating	27	28	20	20 / 7
replication	93	116	34	19 / 74
replications	8	10	5	1 / 7
may	970	1050	99	761 / 209
might	324	345	87	211 / 113
could	328	349	86	195 / 133

All 15,280 retained texts form the denominator: 8,427 Qwen-positive and 6,853 negative. Negative texts are not assigned certainty zero. The three-adverb family appears in 87 texts (90 occurrences, 52 papers), of which 62 are model-positive. Replication forms appear in 304 texts (348 occurrences, 75 papers), with 133 positive and 171 negative. Lowercase modals appear in 1,557 texts (1,744 occurrences, 103 papers), with 1,124 positive and 433 negative. Family any-hit counts can be smaller than sums of individual-word counts. The 87-text adverb coverage is too narrow to corroborate broad certainty stability, and the 171 excluded replication-word texts do not establish filtering errors.

All-text event analyses retain 85 sources (44 criterion-negative, 41 positive) with at least ten texts before and ten at/after the project journal year, independent of whether a cue occurs. Sources with zero hits remain in the analysis. Equal-source post-minus-pre incidence contrasts are -0.23 percentage points $[-1.13, +0.67]$ for the three adverbs, -0.68 $[-2.76, +1.21]$ for replication forms and $+2.57$ $[-1.40, +6.86]$ for lowercase modals. Source-bootstrap intervals use 19,999 draws within criterion groups, seed 20261005, conditional on retrieval, labels, dates and rules. They are pointwise diagnostic intervals, not a selected confirmatory family.

Table S10: Identical-source filter comparison: 72 sources, 40 criterion-negative and 32 positive. Differences and intervals are percentage points.

Observable	Text denominator / contrast	Negative / positive n	Difference, pp	95% source interval, pp
Three adverbs	All text	40 / 32	-0.33	$[-1.25, +0.55]$
Three adverbs	Qwen-positive	40 / 32	-0.40	$[-1.56, +0.68]$
Three adverbs	Filtered minus all	40 / 32	-0.08	$[-0.72, +0.52]$
Replication forms	All text	40 / 32	-0.22	$[-1.87, +1.40]$

Table S10 (continued)

Observable	Text denominator / contrast	Negative / positive n	Difference, pp	95% source interval, pp
Replication forms	Qwen-positive	40 / 32	-1.77	[-3.99, +0.26]
Replication forms	Filtered minus all	40 / 32	-1.55	[-3.03, -0.24]
Lowercase modals	All text	40 / 32	+0.37	[-2.83, +3.90]
Lowercase modals	Qwen-positive	40 / 32	-0.72	[-5.21, +4.11]
Lowercase modals	Filtered minus all	40 / 32	-1.09	[-3.33, +1.31]

The table holds the 72 source papers fixed across all-text and Qwen-positive denominators, separating text selection from source-eligibility changes. For replication-word incidence, the filtering-associated shift is -1.55 points $[-3.03, -0.24]$; resampling observed exact-text-sharing components gives $[-3.09, -0.30]$. The latter uses the existing 89-component source graph and does not model unknown shared citing articles. This descriptive contrast shows that conditional text denominators can change the lexical comparison; it is not a causal filter effect, a semantic acknowledgment rate or independent model-score validation. The scoring instructions already mention some cues and replication discussion.

Of all retained texts, 112 exceed the historical Qwen 700-character input limit and 54 exceed the 1,200-character second-scorer limit. Prefix checks tokenize the complete original text and retain only tokens wholly ending inside the prefix, never manufacturing a cue from a cut token. Adverb-family hit texts fall from 87 to 82 and 84 at these limits; replication-family presence stays 304 although multiplicity changes. Complete wordwise, case/date-exclusion, token-density, recorded-upload-year, omitted-event-year, symmetric-window, conflicting-year and leave-project-out outputs are retained. All frozen word families and completed specifications are retained, including adverse sensitivities; the main emphasis was chosen post hoc.

The added Staunton source has no core-adverb or replication-form hits. Under the 106-source convention, those raw hit counts remain 87 and 304, while the text denominator becomes 15,302. All-text event eligibility becomes 86 papers (44 negative, 42 positive); the replication-word contrast is -0.65 percentage points $[-2.66, +1.20]$. The matched 72-source filter comparison remains exactly unchanged. No 106-source connected-component interval is claimed. This sensitivity does not convert the literal observables into a validated linguistic or psychological construct.

The accompanying additional-diagnostics files retain RULES_FROZEN.json, the implementation clarification, aggregate source-year inputs, all twelve wordwise outputs, all source and component contrasts, and portable analysis/tests. Separate implementations checked the token rules and reproduced the reported statistics. This establishes computational fidelity to the declared rules, not human agreement or construct validity.

S10 Exact selection decomposition and concentration checks

This extension was specified after the original model scores and the matched lexical comparisons were known. Its protocol froze the accounting identities, all three existing families, source/project influence rules, support checks and five time configurations before these new calculations. Promotion of text selection to the article’s main question is itself retrospective. The extension does

not introduce a new decomposition method, semantic labels, model fitting or an independent validation set.

For source i and period t , p^+ and p^- denote retained and excluded literal-family incidence, q the excluded fraction and $d=p^+-p^-$. The retained-minus-all gap is $G=qd$. The symmetric identity $\Delta G=\Delta q(d_{\text{post}}+d_{\text{pre}})/2+\Delta d(q_{\text{post}}+q_{\text{pre}})/2$ is applied to each source, then averaged within criterion groups before the negative-minus-positive subtraction. The first term is labeled retention-share, through the complement of the excluded share; the second is stratum-distribution. They are signed algebraic allocations and may cancel. Neither their signs nor their ratios identify causes or percentages explained. Products of group means are not substituted for source-level products.

The matched set contains 72 sources (40 negative, 32 positive), with 4,812 pre-period and 7,187 post-period text units. Every source has excluded texts in each period, so the decomposable and matched populations coincide. The alternative accounting identity $G=q p^+-H^-/N_{\text{all}}$, where H^- is the excluded-stratum hit count and N_{all} the all-text count, is also verified on full support. Missing excluded-stratum rates are never filled with zero. A stricter minimum of ten excluded texts in each period retains 59 sources (32 negative, 27 positive; Table S13).

The original lexical intervals in Table S10 and main Fig. 3 retain seed 20261005. Main Table 2 and Tables S11–S13 use the later protocol’s seed 20261006: 19,999 joint source-bootstrap draws within criterion groups. The total point estimates match exactly; the slightly different interval endpoints reflect a documented Monte Carlo stream. Known exact-text-sharing components are resampled separately as a dependence sensitivity. These are conditional, pointwise summaries, not multiplicity-adjusted tests or population uncertainty over four scientific fields.

Table S11: Every leave-project-out filter-shift comparison. Values and intervals are percentage points, conditional on the frozen corpus and rule set.

Family	Omitted project	Negative / positive n	Shift	95% source interval
Three adverbs	EERP	36 / 26	-0.22	[-0.93, +0.42]
Three adverbs	ML1	39 / 28	-0.03	[-0.73, +0.60]
Three adverbs	ML3	36 / 30	-0.13	[-0.81, +0.53]
Three adverbs	RPP	9 / 12	+0.28	[-0.59, +1.17]
Replication forms	EERP	36 / 26	-1.65	[-3.27, -0.18]
Replication forms	ML1	39 / 28	-1.72	[-3.27, -0.32]
Replication forms	ML3	36 / 30	-1.66	[-3.29, -0.22]
Replication forms	RPP	9 / 12	-0.70	[-2.39, +0.90]
Lowercase modals	EERP	36 / 26	-1.11	[-3.63, +1.60]
Lowercase modals	ML1	39 / 28	-1.12	[-3.53, +1.45]
Lowercase modals	ML3	36 / 30	-0.88	[-3.25, +1.66]

Table S11 (continued)

Family	Omitted project	Negative / positive n	Shift	95% source interval
Lowercase modals	RPP	9 / 12	-1.06	[-4.46, +2.90]

RPP supplies 51 of 72 matched sources and has a within-project replication-form shift of -2.048 points $[-3.982, -0.344]$. The other project group sizes are EERP 4 negative/6 positive, Many Labs 1 1/4 and Many Labs 3 4/2. The Many Labs 1 replication-form point estimate is $+0.221$ points; no interval is reported for its singleton negative cell. Equal-project and observed pooled-project-share standardizations give -1.018 and -1.716 points, respectively. Their manuscript-facing intervals are suppressed because resampling within project $\times$ criterion strata would keep the singleton fixed. All point estimates and the clearly labeled conditional audit columns remain in `within_project.csv` and `project_standardized.csv`.

No single-source deletion reverses the replication-form shift: the range is -1.714 to -1.188 points. Deleting `rpp.68` changes the estimate most, by $+0.365$ points. The complete per-source results for every family, rather than only influential or favorable cases, are in `leave_source_out.csv`. The non-zero individual-source interval for the small adverb retention component includes zero under known-text-component resampling; this is not a robust isolated finding. Both modal component intervals include zero.

Table S12: All predefined timing configurations for every family. Eligibility is recalculated under the same rule for each configuration; these are not dates of author awareness. Values and intervals are percentage points.

Family	Alignment	Negative / positive n	Shift	95% source interval
Three adverbs	Journal year	40 / 32	-0.08	[-0.72, +0.53]
Three adverbs	Recorded upload year	39 / 32	-0.07	[-0.72, +0.55]
Three adverbs	Omit event year	39 / 31	+0.13	[-0.59, +0.80]
Three adverbs	Five-year windows	35 / 26	+0.27	[-0.35, +0.91]
Three adverbs	Latest text year	40 / 32	+0.00	[-0.67, +0.68]
Replication forms	Journal year	40 / 32	-1.55	[-3.04, -0.21]
Replication forms	Recorded upload year	39 / 32	-1.55	[-3.02, -0.21]
Replication forms	Omit event year	39 / 31	-1.97	[-3.55, -0.53]
Replication forms	Five-year windows	35 / 26	-1.52	[-3.46, +0.13]
Replication forms	Latest text year	40 / 32	-1.55	[-3.04, -0.21]
Lowercase modals	Journal year	40 / 32	-1.09	[-3.36, +1.34]

Table S12 (continued)

Family	Alignment	Negative / positive n	Shift	95% source interval
Lowercase modals	Recorded upload year	39 / 32	-1.22	[-3.50, +1.21]
Lowercase modals	Omit event year	39 / 31	-1.94	[-4.11, +0.28]
Lowercase modals	Five-year windows	35 / 26	-0.50	[-3.37, +2.30]
Lowercase modals	Latest text year	40 / 32	-1.02	[-3.33, +1.48]

Omitting RPP gives a replication-form shift of -0.701 points $[-2.390, +0.904]$, and symmetric five-year windows excluding the event year give -1.523 $[-3.461, +0.131]$. Both intervals span zero, as does the stricter excluded-text support sensitivity below. These are substantive limits on the case’s stability and generality. Known exact-text-component intervals for the main replication-form total, retention and distribution terms are $[-3.084, -0.293]$, $[-0.418, -0.074]$ and $[-2.728, -0.147]$, respectively. They still omit dependence through unidentified citing articles.

Table S13: Minimum ten excluded texts in each period: 59 sources, 32 negative and 27 positive. Each cell is a percentage-point estimate with its conditional 95% source interval.

Family	Total shift	Retention-share term	Distribution term
Three adverbs	+0.10 [-0.40, +0.64]	+0.05 [-0.01, +0.12]	+0.05 [-0.43, +0.55]
Replication forms	-1.44 [-3.13, +0.04]	-0.22 [-0.42, -0.05]	-1.22 [-2.78, +0.15]
Lowercase modals	-0.46 [-2.96, +2.20]	+0.18 [-0.19, +0.54]	-0.64 [-3.11, +2.00]

The source protocol, aggregate inputs, all complete source/project/timing tables and runnable code are in the clean no-new-label selection extension. Separate numerical implementations verified 5,652 reconstruction checks and 292 comparisons with the production outputs and intervals. These checks establish fidelity to the declared identities and resampling scheme, not semantic validity. The bounded external linkage analysis is reported in Section S12; it supplies no independent certainty validation.

S11 Matched artificial-retention reference analysis

The protocol fixed the source set, three pre-existing literal families, inclusion margins, length bands, seed and 19,999 draws before the benchmark calculations, after earlier score, lexical and decomposition outcomes were known. This is a transparent post hoc extension. Each artificial selection keeps the gate’s retained count within source $\times$ period; source $\times$ archived year; or source $\times$ archived year $\times$ length band. Character-length bands are 0–69, 70–350, 351–700, 701–1,200 and at least 1,201. Uniform sampling without replacement is joint across the three marker indicators, preserving their within-stratum overlap. Exact hypergeometric marginal expectations and variances are computed alongside PCG64 simulations, seed 2026100602.

All schemes use 72 sources (40 criterion-negative and 32 positive), 11,999 text units and 6,650 retained positions. The most constrained scheme has 1,677 strata, including 927 permitting nontrivial random selection; 89.4% of texts and 88.4% of retained positions fall in those strata. Of 234 replication-word hit texts in this matched set, 198 are in randomizable strata. There are 124 strata with both marker and selection variation.

Table S14: All nine retention references. Values are percentage points. Envelopes describe artificial selections of fixed archived texts; they are not confidence intervals from resampling source papers.

Family	Matching	Observed	Expected	Residual	95% envelope
Adverbs	Volume	-0.076	0.000	-0.076	[-0.617, +0.611]
Replication	Volume	-1.553	0.000	-1.553	[-1.112, +1.131]
Modals	Volume	-1.093	0.000	-1.093	[-2.400, +2.367]
Adverbs	+ Year	-0.076	-0.255	+0.179	[-0.739, +0.242]
Replication	+ Year	-1.553	-0.503	-1.050	[-1.430, +0.417]
Modals	+ Year	-1.093	-0.246	-0.848	[-2.352, +1.851]
Adverbs	+ Length	-0.076	-0.220	+0.144	[-0.680, +0.251]
Replication	+ Length	-1.553	-0.651	-0.902	[-1.542, +0.242]
Modals	+ Length	-1.093	-0.272	-0.821	[-2.253, +1.733]

The replication-word observation lies only 0.011 points below the simulated lower endpoint in the year-and-length scheme. An explicitly post-outcome numerical precision check gives a binomial/order-statistic Monte Carlo bracket of $[-1.559476, -1.523360]$ for that endpoint, containing the observed -1.552904 . Brackets for all 18 endpoints are retained. This numerical diagnostic changes no scientific specification and is not an outcome hypothesis test. Adverb and modal observations lie inside all three envelopes.

The most constrained residual stays negative after each single-project omission (approximately -0.82 to -1.07 points), while within-project residuals vary: EERP -2.32 , Many Labs 1 $+1.81$, Many Labs 3 -0.31 and RPP -1.12 . The negative cell in Many Labs 1 has one source, so these figures do not support inference to project populations. Within criterion-negative sources the all-text change is $+0.359629$ points, the year-and-length reference change is -0.520255 , and actual retained change is -1.169599 . Composition matching therefore already reproduces the within-group direction reversal.

Online Resource 3 contains text-free per-context inputs, joint-pattern stratum counts, all source and group incidences, every project concentration check and complete numerical outputs. A standalone replay reproduces the six analysis CSVs byte for byte. Independent numerical verification reproduces all nine original numerical files and 114,764 checks. Existing source-bootstrap intervals are stored separately from the artificial-retention envelopes to prevent interpretation as the same uncertainty calculation.

S12 External article linkage and descriptive retention

A separate fixed, source-based convenience pilot examined whether archived citation contexts could be linked to independently produced article-level valence judgments from Hardwicke et al. (2021). The source archive (<https://osf.io/gyzbm/>) and codebook define one adjudicated

nominal judgment per citing article–target case, rather than per sentence. The recovery sample and matching rules were fixed before acquisition. The eight eligible article–case pairs and twelve context identifiers, allowed label fields and descriptive analysis script were recorded before the selected labels and existing gate decisions were inspected. This local sequence is not a preregistration of the study.

The fixed sample contained 12 article–case rows from 10 citing DOIs. Public full text was recovered for 10 rows from eight articles. Complete archived-context matches within exact target-citing paragraphs yielded 12 eligible contexts across eight article–case pairs from seven distinct citing articles; one DOI appears in both target cases. These contexts cover 11 of 21 recovered target-anchored paragraph–case units. Three other complete-context body matches were outside the target-citing paragraphs and were rejected. Two article documents remained inaccessible and were not replaced. All eligible matches agreed with the external publication year, were unique within the recovered candidates, and had no detected normalized development-text overlap.

Table S15: Retention of linked archived contexts by external article-level valence. Counts concern an availability-conditioned subset. Empty valence categories have no estimable retention rate. Article-level judgments are not sentence-level gold labels.

Target case	Article valence	Pairs	Contexts	Retained	Retention
Carter	Favorable	2	4	4	4/4
Carter	Unfavorable	0	0	0	Not estimable
Carter	Equivocal	1	1	1	1/1
Carter	Unclassifiable	0	0	0	Not estimable
Caruso	Favorable	4	6	6	6/6
Caruso	Unfavorable	0	0	0	Not estimable
Caruso	Equivocal	1	1	0	0/1
Caruso	Unclassifiable	0	0	0	Not estimable

All eight selected article–case rows were included under the original source’s exclusion field. Six favorable pairs supplied ten linked contexts, all retained; two equivocal pairs supplied two contexts, one retained. No unfavorable or unclassifiable pair was present. The excluded Caruso context is therefore not a verified dropped criticism, and differential coverage of critical citations cannot be estimated. The eight pairs form seven citing-article clusters; no confidence interval, hypothesis test, accuracy estimate or threshold selection is reported.

The source CSV uses favourable. A separately recorded post-unblinding amendment mapped favourable to favorable and unfavourable to unfavorable as spelling variants of the same nominal categories; no unfavorable value was observed. Equivocal and unclassifiable remained unchanged. Original strings and the strict initial output are preserved in Online Resource 4. No sample, linkage, exclusion rule, denominator, model output or category meaning changed.

The original human coding spans are not released, so recovered target-citing paragraphs cannot reconstruct those exact annotation units. Current-version matches also do not authenticate the original ingestion version, and exact-text overlap checks cannot establish article-level independence across unknown versions. These results describe provenance and frozen-gate retention among linked archived excerpts. They neither independently validate certainty or assertion accuracy nor

support causal or population inference. Online Resource 4 supplies the complete 12-row recovery ledger, eight original article-level labels, twelve text-free gate rows, all category summaries, provenance hashes and a standard-library numerical replay. Human annotation data retain the source archive’s CC BY 4.0 attribution; publisher full text and private annotation metadata are excluded.

S13 Reproduction and scope boundaries

The repaired source-paper estimates were checked using an independent annual-sum and weighted-design-matrix implementation. All 96 level, 85 lifetime-slope and 72 event estimates agree to floating-point tolerance. Original archive replay, component-score means, input hashes, all finite label allocations and complete sensitivity tables are retained. Measurement code was checked against independent formulas for agreement and tied-rank calculations. These checks validate implementation and reconstruction, not the intended psychological construct.

Online Resources 2 and 3 provide numerical inputs, source ledgers, analysis code, aggregate measurement derivations, complete outputs and file hashes. They exclude private annotation-export metadata and participant-level primary-study files. Third-party texts and data retain their original source terms. The files are supplied as supplementary material; no new public repository release is represented. Online Resource 4 contains the separate external-linkage pilot inputs, full recovery ledger, descriptive outputs and replay.

No independently adjudicated target-proposition linkage or independent certainty-validation result was established for the main archive. Consequently, the study remains limited to the visible word-form and archived algorithmic outcomes defined in the manuscript. The absence of such a result does not undo the achieved internal model–human rank-agreement criterion. Stronger semantic or target-response claims would require evidence specifically designed for those constructs.